

# Aman Syed

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## PERSONAL PROFILE

**Data Scientist & AI Engineer** with 2+ years delivering production NLP systems, GenAI insights products, and LLM evaluation pipelines on AWS. Reduced manual analysis time by **95%** through a multi-agent media intelligence platform; improved sentiment model weighted F1 to **71%** by evaluating 33+ prompt configurations across 6 LLMs; deployed a 15-endpoint AI narrative platform generating automated client briefings daily across 16 modules. Works across the full data science lifecycle – model evaluation, prompt engineering, fine-tuning benchmarks, and production monitoring. **Core stack: Python, PyTorch, OpenAI, Gemini, Claude, FastAPI, PostgreSQL, SageMaker.**

## EXPERIENCE

### Associate Data Scientist

*Oct'24 – Present*

*Penta Group, London*

- **Multi-Agent Transcript Platform:** Reduced manual analysis time by **95%** (20 hrs to under 5 min) via multi-agent NLP platform for text, audio, and video using RevAI speech-to-text, 3-LLM consensus speaker extraction (Gemini/OpenAI/Claude), and a 3-stage quote pipeline; relevance filtering, subquote precision extraction, and quality evaluation with composite re-ranking – achieving **92% precision**.
- **Sentiment Model Optimisation:** Improved model to **71% weighted F1** on 1,000 human-labelled articles by evaluating **33+ prompt configurations** across 6 LLMs; benchmarked Llama 3.1/3.2 fine-tuning vs. zero-shot; built statistical drift monitoring with p-value alerting and per-class F1/MAE tracking against a 12,791-row gold dataset.
- **AI Insights Narrative Platform:** Deployed 15-endpoint FastAPI platform generating sentiment-driven narratives across **16 client-facing modules** used in daily briefings; engineered async embeddings, KMeans/HDBSCAN clustering with MMR diversity, primary/fallback LLM architecture, and Parquet embedding caching – validated by **1,245 automated tests**.
- **Conversational RAG System:** Developed SSE-streaming RAG with message history context engineering and adaptive context windowing (entity/keyword anchors) – achieving **62% context reduction** without quality degradation; integrated adaptive LLM routing, query enhancement pipeline, NMF topic diversification, and structured inline citations – **cutting LLM costs by 60%**.
- **Hybrid Search Engine:** Engineered vector (pgvector HNSW), full-text (tsvector), and NER entity search with Reciprocal Rank Fusion across **6 dynamic weight tiers**; implemented two-tier cache (exact SHA256 hash + semantic similarity at 0.95 threshold, 1-hour TTL) and per-user token bucket rate limiting at 60 RPM.
- **GPU-Accelerated NLP Pipeline:** Scaled to process **2–3 million daily articles** via SQS-driven ECS workers running GLiNER NER and Gemma-300m embeddings on NVIDIA GPUs; reduced redundant inference via 30-day SHA256 deduplication across daily-partitioned PostgreSQL/pgvector tables on AWS ECS.
- **Newsletter Summarisation Pipeline:** Automated client newsletter curation processing **2,000+ articles per portfolio** using multi-provider LLM architecture (GPT-4o/5, Gemini 2.5, Claude 4) with prompt optimisation framework and multi-model A/B testing.
- **Storyline Identification Service:** Built Dockerized GPT-4o pipeline with OpenAI embeddings, UMAP dimensionality reduction, and HDBSCAN clustering to extract and summarise narratives from up to **10,000 articles**; supported organic, guided, and user-defined discovery with versioned JSONL prompt templates.
- **LLM Evaluation Framework:** Benchmarked model quality across BLEU, ROUGE, BERTScore, semantic similarity, self-BLEU, entropy, and type-token ratio; evaluated GPT-4.1, Gemini 2.5 Pro/Flash, Llama 3.1/3.2, and Claude to drive model upgrade decisions and reduce per-inference cost.
- **Inference Optimisation:** Achieved **3x throughput** and sub-500ms latency deploying BERT-based models as serverless SageMaker endpoints with dynamic batching; built dual LDA/NMF topic extraction pipeline with automatic model selection and interactive network/treemap visualisations.
- **MCP Server:** Built Model Context Protocol server exposing **5 AI-client tools** (hybrid search, vector search, full-text search, entity search, RAG Q&A) over SSE with JWT authentication; enables external AI agents to query the article knowledge base with cited answers via stateless, config-driven endpoints.
- **Productionized AI Scraper:** Adopted by engineering team for structured data ingestion; evaluated CSS selector generation across LLMs with **9-step HTML cleaning** (70–80% size reduction) and 4-tier validation pipeline.
- **Infrastructure & DevOps:** Delivered **5x faster file ingestion** via chunked-transfer uploads over parallel

WebSockets; built CI/CD across integration, pilot, and production using CodeBuild/CodePipeline; migrated hardcoded credentials to AWS SSM Parameter Store across all environments.

## Financial Research Assistant

Sep'24 – Oct'24

The University of Edinburgh – Business School, Edinburgh

- **Developed Python data pipeline using Refinitiv API** to collect financial data across **500+ U.S. companies** spanning 3 fiscal years for environmental research, automating previously manual data extraction.
- **Compiled earnings call transcripts and presentation briefs** from Refinitiv Workspace, building a structured dataset for text and sentiment analysis of corporate environmental disclosures.

## Data Engineer Intern

Dec'22 – Aug'23

C.H. Robinson, Mumbai

- **Serverless AWS Pipeline:** Designed & implemented a data pipeline (Lambda, S3, API Gateway) processing **100K daily API records** to improve data availability and system reliability.
- **Workflow Automation:** Automated processes using Selenium & Python scripts, saving **90+ monthly engineering hours** and doubling data capture capacity.
- **Logistics Reporting:** Automated daily client report generation with Python, increasing output to **20+ reports daily** and boosting client satisfaction.
- **Lambda Optimisation:** Optimised 200+ AWS Lambda functions through SQL query tuning and Python code refactoring, improving data processing efficiency by **40%**.

## PROJECTS

### Finance Flow (frontend · backend)

2024 – Present

React 19, TypeScript, Tailwind CSS, FastAPI, Google Sheets API, Render

- Built full-stack personal finance suite: React 19 dashboard with analytics, budgets, dark mode, and Google Sheets sync; Python FastAPI backend using the Strategy Pattern to auto-detect and normalise CSVs from 6 UK banks (Amex, Revolut, Monzo, Starling, BoS) into a unified schema with Frankfurter exchange rate proxy. Deployed frontend on GitHub Pages, backend on Render.

### Knock Knock Physio (source)

2024

React, TypeScript, Vite, GitHub Pages

- Launched a production business website for a mobile physiotherapy practice; online booking, WhatsApp integration, pricing plans, testimonials, and protected admin dashboard.

## TECHNICAL SKILLS

- **Languages:** Python, SQL, TypeScript, JavaScript, R, Java
- **LLMs & Gen AI:** OpenAI GPT, Google Gemini, Anthropic Claude, RAG, LLM Fine-tuning, Prompt Engineering, HuggingFace Transformers, Multi-agent Orchestration, MCP
- **ML/NLP:** PyTorch, SpaCy, BERT, GLiNER, Gensim, Scikit-learn, Pandas, NumPy, LDA/NMF, Sentence Embeddings, UMAP, HDBSCAN
- **Backend, DBs & APIs:** FastAPI, Flask, Pydantic, PostgreSQL (pgvector), MySQL, MongoDB, REST APIs, SSE
- **Cloud (AWS):** SageMaker, ECS, Lambda, S3, SQS, API Gateway, CloudWatch, SSM Parameter Store, ECR, CodeBuild, CodePipeline
- **Frontend & Web:** React, Vite, TailwindCSS, Node.js, ipywidgets
- **MLOps & DevOps:** Docker, Git, CI/CD, GitHub Actions, pytest, Inference Optimisation, Agile/Scrum
- **Other:** Vertex AI, Google ADK, Selenium, Refinitiv Workspace, Postman, Matplotlib, Plotly, Power BI

## EDUCATION

### MSc Statistics with Data Science

Sep'23 – Aug'24

The University of Edinburgh, Edinburgh

2:2 Pass

### BEng Computer Engineering

Aug'19 – Jun'23

University of Mumbai, Mumbai

9.83/10 CGPA

## CERTIFICATIONS

- Architecting with Google Compute Engine Specialization – Google Cloud
- Machine Learning Foundations: A Case Study Approach – University of Washington